# MT-1: Black Hole Entropy and Four Laws of Thermodynamics in Spectral Causal Theory

Wald Entropy, Logarithmic Corrections, and Modified Hawking Radiation from the Nonlocal Spectral Action

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**Abstract**

We derive the complete black hole entropy formula from the one-loop SCT spectral action on a Schwarzschild background. The Wald entropy formula, applied to the spectral action Lagrangian containing the Einstein–Hilbert term and two nonlocal curvature-squared sectors, yields three distinct contributions:

$$S = \frac{A}{4G} + \frac{13}{120\pi} + \frac{37}{24}\ln\Big(\frac{A}{\ell_P^2}\Big) + \mathcal{O}(1).$$

The first term is the Bekenstein–Hawking area law from the Einstein–Hilbert sector. The second is a topological correction from the Weyl-squared sector, computed via the Jacobson–Myers formula with coefficient $\alpha_C = 13/120$ established in NT-1b Phase 3. The third is the one-loop logarithmic correction with $c_{\log} = 37/24$, a *parameter-free* prediction determined entirely by the Standard Model field content plus the graviton sector via the Sen (2012) formula. The $R^2$ sector contributes zero on Schwarzschild ($R = 0$).

We verify all four laws of black hole thermodynamics in the SCT framework: the zeroth law (constancy of surface gravity), the first law (Iyer–Wald theorem), the third law (extremal limit), and the second law—which is **conditional** on the MR-2 ghost resolution (Wall's linearized stability condition). Modified Hawking radiation due to the nonlocal form factors is shown to be negligible for all astrophysical black holes ($T_H/\Lambda \sim 10^{-10}$). The SCT prediction $c_{\log} = +37/24$ has the *opposite sign* from the loop quantum gravity result $c_{\log}^{\mathrm{LQG}} = -3/2$, providing a potential observational discriminant.

## Contents

**1 Introduction** **2**
1.1 Context and Motivation . . . 2
1.2 Relationship to Previous Results . . . 3
1.3 Notation and Conventions . . . 3

**2 The Spectral Action on Schwarzschild Backgrounds** **3**
2.1 The One-Loop Effective Action . . . 3
2.2 Simplification on Schwarzschild . . . 3

**3 Wald Entropy of the Spectral Action** **4**
3.1 The Wald–Noether Charge Formula . . . 4
3.2 Einstein–Hilbert Sector . . . 4

**4 Topological Correction from the Weyl Sector** **4**
4.1 The Jacobson–Myers Formula . . . 4
4.2 Application to SCT . . . 4
4.3 Jacobson–Myers Consistency Check . . . 5
4.4 $R^2$ Sector Vanishes . . . 5

**5 Logarithmic Correction $c_{\log} = 37/24$** **5**
5.1 The Sen Formula . . . 5
5.2 Standard Model Field Content . . . 5
5.3 Decomposition and Result . . . 5

**6 Full Entropy Formula** **6**
6.1 Hierarchy of Contributions . . . 6

**7 Four Laws of Black Hole Thermodynamics in SCT** **6**
7.1 Zeroth Law: Constancy of Surface Gravity . . . 6
7.2 First Law: Energy Balance . . . 7
7.3 Second Law: Generalized Area Theorem . . . 7
7.4 Third Law: Extremal Limit . . . 7
7.5 Summary of Four Laws . . . 8

**8 Modified Hawking Radiation** **8**
8.1 Standard Hawking Temperature . . . 8
8.2 SCT Greybody Modification . . . 8
8.3 Threshold Mass . . . 8

**9 Comparison with LQG, String Theory, and Asymptotic Safety** **8**

**10 Summary** **9**
10.1 Downstream Tasks . . . 9

# 1 Introduction

## 1.1 Context and Motivation

Black hole thermodynamics, initiated by Bekenstein [1] and Hawking [2], remains one of the deepest connections between gravity, quantum mechanics, and statistical physics. The Bekenstein–Hawking entropy $S = A/(4G)$ is a universal prediction of classical general relativity, while quantum corrections—particularly the logarithmic correction $c_{\log} \ln\big(A/\ell_P^2\big)$—encode the ultraviolet structure of the gravitational theory and can discriminate between competing approaches to quantum gravity.

In the Spectral Causal Theory (SCT) framework, the one-loop effective action on an arbitrary curved background takes the form

$$S[g] = S_{\mathrm{EH}}[g] + S_{\mathrm{quad}}[g] + S_{\mathrm{matter}}[g], \tag{1}$$

where the curvature-squared sector $S_{\mathrm{quad}}$ arises from the spectral action principle $\mathrm{Tr}\, f(D^2/\Lambda^2)$. The form factors $F_1(\Box/\Lambda^2)$ (Weyl sector) and $F_2(\Box/\Lambda^2, \xi)$ (Ricci-scalar sector), with SM-summed coefficients $\alpha_C = 13/120$ and $\alpha_R(\xi) = 2(\xi - 1/6)^2$, were established in NT-1b Phase 3. The full nonlinear field equations were derived in NT-4b, including the Wald entropy variation needed for the present calculation.

### 1.2 Relationship to Previous Results

This derivation depends on and extends:

- **NT-1b Phase 3:** The SM-summed form factor coefficients $\alpha_C = 13/120$ (parameter-free), $\alpha_R(\xi) = 2(\xi - 1/6)^2$.
- **NT-2:** The entire-function property of $F_1(z)$, $F_2(z, \xi)$, ensuring no spurious poles in the action.
- **NT-4b:** The full nonlinear field equations and the Wald entropy variation $\delta S_{\text{spec}}/\delta R_{\mu\nu\rho\sigma}$.
- **MR-2:** Ghost catalogue and unitarity conditions (conditional input for the second law).

### 1.3 Notation and Conventions

| Symbol | Definition |
|---|---|
| $g_{\mu\nu}$ | Metric tensor (Lorentzian signature $(-,+,+,+)$) |
| $R_{\mu\nu}$ | Ricci tensor |
| $R$ | Ricci scalar |
| $C_{\mu\nu\rho\sigma}$ | Weyl tensor |
| $G_{\mu\nu}$ | Einstein tensor |
| $\kappa^2$ | $8\pi G$ |
| $\Lambda$ | Spectral cutoff scale |
| $\ell_P = \sqrt{\hbar G/c^3}$ | Planck length |
| $M_{\text{Pl}}$ | Planck mass |
| $\chi(\mathcal{H})$ | Euler characteristic of the horizon cross-section |
| $\hat{\epsilon}_{\mu\nu}$ | Binormal to the bifurcation surface |

## 2 The Spectral Action on Schwarzschild Backgrounds

### 2.1 The One-Loop Effective Action

The curvature-squared sector of the SCT spectral action is

$$S_{\text{quad}} = \frac{1}{16\pi^2} \int \mathrm{d}^4x \sqrt{-g} \left[\alpha_C \, F_1(\Box/\Lambda^2) \, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} + \alpha_R(\xi) \, F_2(\Box/\Lambda^2, \xi) \, R^2\right], \tag{2}$$

where $\alpha_C = 13/120$ and $\alpha_R(\xi) = 2(\xi - 1/6)^2$.

### 2.2 Simplification on Schwarzschild

On a Schwarzschild background with mass $M$:

$$R = 0, \tag{3}$$

$$R_{\mu\nu} = 0 \quad \text{(Ricci-flat)}, \tag{4}$$

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} \quad \text{(Weyl = Riemann)}. \tag{5}$$

As a consequence:

1. The $R^2$ sector of $S_{\text{quad}}$ vanishes identically.
2. The Weyl sector reduces to the Riemann-squared sector.
3. The Gauss–Bonnet combination $E_4 = C^2$ on Ricci-flat backgrounds, making the Jacobson–Myers formula directly applicable.

**[Verified]** Equations (3)–(5) verified numerically for the Schwarzschild metric at 7 test radii. The Kretschner scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6$ matches the Weyl invariant $C^2$ to 100-digit precision.

# 3 Wald Entropy of the Spectral Action

## 3.1 The Wald–Noether Charge Formula

The Wald entropy [4] for a diffeomorphism-invariant Lagrangian $\mathcal{L}(g_{\mu\nu}, R_{\alpha\beta\gamma\delta})$ is

$$S_{\text{Wald}} = -2\pi \oint_{\mathcal{H}} \frac{\partial \mathcal{L}}{\partial R_{\mu\nu\rho\sigma}} \hat{\epsilon}_{\mu\nu}\hat{\epsilon}_{\rho\sigma}\, \mathrm{d}A, \tag{6}$$

where $\hat{\epsilon}_{\mu\nu}$ is the binormal to the bifurcation surface $\mathcal{H}$, normalized as $\hat{\epsilon}_{\mu\nu}\hat{\epsilon}^{\mu\nu} = -2$ (Lorentzian).

## 3.2 Einstein–Hilbert Sector

For $\mathcal{L}_{\text{EH}} = R/(16\pi G)$:

$$\frac{\partial \mathcal{L}_{\text{EH}}}{\partial R_{\mu\nu\rho\sigma}} = \frac{g^{\mu[\rho}g^{\sigma]\nu}}{16\pi G}. \tag{7}$$

The binormal contraction gives $g^{\mu[\rho}g^{\sigma]\nu}\hat{\epsilon}_{\mu\nu}\hat{\epsilon}_{\rho\sigma} = -2$, so

$$S_{\text{EH}} = \frac{A}{4G}, \tag{8}$$

recovering the Bekenstein–Hawking entropy.

**[Verified]** $S_{\text{EH}}(10\, M_\odot) = 1.0495 \times 10^{79}$. Computed at 100-digit precision using CODATA 2022 constants.

# 4 Topological Correction from the Weyl Sector

## 4.1 The Jacobson–Myers Formula

On Ricci-flat backgrounds, $C^2 = \text{Riem}^2 = E_4$ (the Gauss–Bonnet density), so the Weyl-squared contribution to the Lagrangian can be treated as a Lovelock-type term. The Jacobson–Myers (JM) formula [6] gives the entropy of the Gauss–Bonnet density:

$$\Delta S_{E_4} = 8\pi\gamma\, \chi(\mathcal{H}), \tag{9}$$

where $\gamma$ is the coefficient of the Gauss–Bonnet term and $\chi(\mathcal{H})$ is the Euler characteristic of the horizon cross-section.

## 4.2 Application to SCT

In the SCT action (2), evaluated in the local limit $F_1(\Box/\Lambda^2) \to F_1(0) = 1$ on the horizon, the effective Gauss–Bonnet coefficient is

$$\gamma = \frac{\alpha_C}{16\pi^2} = \frac{13/120}{16\pi^2}. \tag{10}$$

For a Schwarzschild black hole, the horizon is topologically $S^2$ with $\chi(S^2) = 2$. Therefore:

$$\boxed{\Delta S_{\text{Weyl}} = 8\pi \frac{\alpha_C}{16\pi^2} \cdot 2 = \frac{\alpha_C}{\pi} = \frac{13}{120\pi} \approx 0.03448.} \tag{11}$$

This is a **topological** correction: it depends on the topology of the horizon ($\chi = 2$) but not on its area.

**Remark 4.1** (Independence from area)**.** The Weyl correction $\Delta S_{\text{Weyl}} = 13/(120\pi)$ is a constant, independent of the black hole mass or horizon area. This reflects its origin as a topological (Euler characteristic) contribution, not a geometric one.

**Remark 4.2** (Kerr generalization)**.** For Kerr black holes, the horizon remains topologically $S^2$ with $\chi = 2$, so the Weyl correction is identical. For hypothetical toroidal horizons ($\chi = 0$), the correction would vanish.

### 4.3 Jacobson–Myers Consistency Check

The JM formula can also be written as

$$\Delta S_{\mathrm{JM}} = 32\pi^2 \gamma\, \chi(\mathcal{H}) \cdot \frac{1}{4\pi}, \tag{12}$$

which reduces to $\Delta S_{\mathrm{JM}} = \alpha_C\, \chi(\mathcal{H})/(2\pi)$. For $\chi = 2$: $\Delta S_{\mathrm{JM}} = \alpha_C/\pi = 13/(120\pi)$.

**[Verified]** JM formula consistency: $|\Delta S_{\mathrm{JM}} - \Delta S_{\mathrm{simplified}}| = 4.5 \times 10^{-103}$ at 100-digit mpmath precision. The alternative Noether charge computation (direct integration of the Weyl tensor binormal contraction on the Schwarzschild horizon) agrees after accounting for the Gauss–Bonnet decomposition terms.

### 4.4 $R^2$ Sector Vanishes

The $R^2$ sector of the Wald entropy involves $\partial(R^2)/\partial R_{\mu\nu\rho\sigma}$, which is proportional to $R$. On Schwarzschild, $R = 0$, so:

$$\Delta S_{R^2}\big|_{\mathrm{Sch}} = 0 \quad \text{(exact)}. \tag{13}$$

## 5 Logarithmic Correction $c_{\log} = 37/24$

### 5.1 The Sen Formula

The one-loop logarithmic correction to the Bekenstein–Hawking entropy for a non-extremal black hole is given by the Sen (2012) formula [7]:

$$c_{\log} = \frac{1}{180}\big[2\, N_s + 7\, N_F - 26\, N_V + 424\big], \tag{14}$$

where $N_s$ is the number of real scalars, $N_F$ the number of Dirac fermions, $N_V$ the number of gauge bosons, and the $+424$ is the graviton contribution.

**[Important note]** The coefficient $-26$ for vectors arises from the proper vector determinant ($+62$) minus two Faddeev–Popov ghosts ($2 \times 44 = 88$), giving $62 - 88 = -26$ per vector. The fermion coefficient is $+7$ per *Dirac* fermion (not $+7/2$ per Weyl). The literature review (MT1-LR) corrected the original extraction which had used $N_F = 45$ Weyl instead of $N_F = 22.5$ Dirac, changing $c_{\log}$ from $211/180$ to $37/24$.

### 5.2 Standard Model Field Content

From NT-1b Phase 3 (CPR [13]):

| Field | Spin | Count | Convention |
|---|---|---|---|
| Real scalars (Higgs) | 0 | $N_s = 4$ | Real components of doublet |
| Dirac fermions | 1/2 | $N_F = 22.5$ | $N_f = 45$ Weyl $\to$ 45/2 Dirac |
| Gauge bosons | 1 | $N_V = 12$ | SU(3)×SU(2)×U(1) |

### 5.3 Decomposition and Result

$$\text{Scalars:} \quad 2 \times 4 = 8, \tag{15}$$

$$\text{Fermions:} \quad 7 \times 22.5 = 157.5, \tag{16}$$

$$\text{Vectors:} \quad -26 \times 12 = -312, \tag{17}$$

$$\text{Graviton:} \quad +424. \tag{18}$$

The grand total is $8 + 157.5 - 312 + 424 = 277.5$, so

$$\boxed{c_{\log} = \frac{277.5}{180} = \frac{37}{24} \approx 1.542.} \tag{19}$$

**Remark 5.1** (Sign analysis)**.** The matter contribution alone is $8 + 157.5 - 312 = -146.5$ (negative, dominated by the vector Faddeev–Popov ghosts). The graviton contribution $+424$ overwhelms this, making the total *positive*. Thus quantum corrections *increase* the entropy beyond the classical area law.

**Remark 5.2** (Pure gravity cross-check)**.** For pure gravity (no matter fields): $c_{\log}^{\text{pure}} = 424/180 = 106/45 \approx 2.356 > 0$. This matches Sen (2012) [7].

**[Verified]** $c_{\log} = 37/24$ verified at exact rational precision via Python `Fraction` arithmetic: `Fraction`$(555, 2)/180 =$ `Fraction`$(37, 24)$. Also verified at 100-digit mpmath precision with $|c_{\log} - 37/24| = 0$ (exact match). Independent CAS cross-check (SymPy × Fraction × mpmath): PASS.

# 6 Full Entropy Formula

Combining the three contributions:

**Theorem 6.1** (SCT Schwarzschild BH entropy)**.** *The entropy of a Schwarzschild black hole of mass $M$ in the SCT framework is*

$$\boxed{S = \underbrace{\frac{A}{4G}}_{\textit{Bekenstein–Hawking}} + \underbrace{\frac{13}{120\pi}}_{\textit{Weyl (topological)}} + \underbrace{\frac{37}{24}\ln\!\left(\frac{A}{\ell_P^2}\right)}_{\textit{one-loop logarithmic}} + \mathcal{O}(1),} \tag{20}$$

*where $A = 16\pi G^2 M^2/c^4$ is the horizon area and $\ell_P^2 = \hbar G/c^3$ is the Planck area.*

## 6.1 Hierarchy of Contributions

For astrophysical black holes, the three contributions satisfy a strict hierarchy:

$$S_{\text{BH}} \gg \Delta S_{\log} \gg \Delta S_{\text{Weyl}}. \tag{21}$$

| $M/M_\odot$ | $S_{\text{BH}}$ | $\Delta S_{\log}$ | $\Delta S_{\text{Weyl}}$ |
|---|---|---|---|
| 1 | $1.05 \times 10^{77}$ | 275.5 | 0.0345 |
| 10 | $1.05 \times 10^{79}$ | 282.6 | 0.0345 |
| 100 | $1.05 \times 10^{81}$ | 289.7 | 0.0345 |
| $10^6$ (Sgr A*-class) | $1.05 \times 10^{89}$ | 318.1 | 0.0345 |
| $4.3 \times 10^6$ (Sgr A*) | $1.94 \times 10^{90}$ | 322.6 | 0.0345 |

**[Verified]** Hierarchy verified for 5 masses from $1\,M_\odot$ to $4.3\times10^6\,M_\odot$. In all cases, $S_{\text{BH}}/\Delta S_{\log} > 10^{74}$ and $\Delta S_{\log}/\Delta S_{\text{Weyl}} > 7000$. The Weyl correction is constant (topological) while the logarithmic correction grows as $\ln M^2$.

# 7 Four Laws of Black Hole Thermodynamics in SCT

## 7.1 Zeroth Law: Constancy of Surface Gravity

**Proposition 7.1** (Zeroth law)**.** *The surface gravity $\kappa$ is constant on the event horizon.*

For Schwarzschild, $\kappa = c^4/(4GM)$ is manifestly constant on the horizon ($S^2$ symmetry). SCT corrections to the background geometry are of order $(\Lambda/M_{\text{Pl}})^2 \sim 10^{-62}$, negligibly small. **Status: PASS.**

### 7.2 First Law: Energy Balance

**Proposition 7.2** (First law — Iyer–Wald)**.** *For any diffeomorphism-invariant Lagrangian, the Wald entropy $S_{\text{Wald}}$ satisfies the first law:*

$$\mathrm{d}M = \frac{\kappa}{2\pi}\,\mathrm{d}S_{\text{Wald}} + \Omega_H\,\mathrm{d}J + \Phi_H\,\mathrm{d}Q. \tag{22}$$

This follows from the Iyer–Wald theorem [5] (Theorem 1), which establishes the first law for any diffeomorphism-invariant Lagrangian with the entropy defined as the Noether charge. Since the SCT spectral action is manifestly diffeomorphism-invariant, the first law holds *exactly* without any additional conditions.

Numerical verification: finite-difference computation of $\mathrm{d}S/\mathrm{d}M$ agrees with the analytic prediction $\mathrm{d}S/\mathrm{d}M = 8\pi GM/(\hbar c)$ with relative error $6.2\times10^{-6}$ (limited by step-size in the finite difference).

**Status: PASS.**

**[Verified]** First law relative error $= 6.22\times10^{-6}$. Surface gravity consistency: $\kappa$ recovered from $\mathrm{d}S/\mathrm{d}M$ matches direct computation to machine precision (error $= 0$ after DR bug fix).

### 7.3 Second Law: Generalized Area Theorem

**Proposition 7.3** (Second law — Wall conditions)**.** *The generalized second law $\mathrm{d}S_{\text{gen}} = \mathrm{d}S_{\text{Wald}} + \mathrm{d}S_{\text{outside}} \geq 0$ holds if the following three conditions are satisfied (Wall [8]):*

1. ***Well-posed initial value problem:*** *the nonlocal equations of motion admit a well-posed Cauchy problem.*
2. ***Null energy condition:*** *$T_{\mu\nu}k^\mu k^\nu \geq 0$ for null vectors $k^\mu$.*
3. ***Linearized stability:*** *the linearized perturbation equations around the black hole background have no exponentially growing modes.*

**Assessment of conditions:**

1. **Well-posedness:** Suggested by NT-2 (entire-function form factors imply no spurious poles in the propagator) and MR-3 ($v_{\text{front}} = c$, macrocausality preserved). However, well-posedness of the full nonlocal system has not been proven rigorously. **Status: SUGGESTED.**

2. **Null energy condition:** Standard Model matter satisfies the NEC classically. Quantum violations (Casimir effect, etc.) are suppressed for macroscopic horizons. **Status: PASS.**

3. **Linearized stability:** The MR-2 ghost catalogue identifies ghost poles in $\Pi_{\text{TT}}(z)$ (two real, three Lee–Wick pairs). The physical ghost at $z_L$ requires resolution via the fakeon prescription (KK analysis) or dark-matter interpretation. If ghosts produce exponentially growing perturbations, the second law would be violated. **Status: CONDITIONAL on MR-2 ghost resolution.**

**Status: CONDITIONAL** — the second law depends on MR-2 ghost resolution. If ghosts are resolved via the fakeon prescription, the effective theory satisfies Wall's conditions in the macroscopic regime, and the generalized second law holds.

### 7.4 Third Law: Extremal Limit

**Proposition 7.4** (Third law)**.** *The surface gravity $\kappa$ cannot be reduced to zero by any finite physical process.*

For Schwarzschild, $\kappa \to 0$ only as $M \to \infty$. For Kerr–Newman, the extremal condition $a^2 + Q^2 = M^2$ receives SCT corrections of order $(\Lambda/M_{\text{Pl}})^2 \sim 10^{-62}$, which are negligible.

**Status: PASS.**

### 7.5 Summary of Four Laws

| Law | Status | Key Evidence |
|---|---|---|
| Zeroth | PASS | $\kappa = \text{const}$ by $S^2$ symmetry |
| First | PASS | Iyer–Wald theorem; numerical: $\delta = 6.2 \times 10^{-6}$ |
| Second | CONDITIONAL | Depends on MR-2 ghost resolution (Wall stability) |
| Third | PASS | $\kappa \to 0$ only as $M \to \infty$; SCT corrections $\sim 10^{-62}$ |

## 8 Modified Hawking Radiation

### 8.1 Standard Hawking Temperature

The Hawking temperature of a Schwarzschild black hole is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}. \tag{23}$$

For a $10\,M_\odot$ black hole: $T_H = 6.17 \times 10^{-9}\,\mathrm{K} = 5.32 \times 10^{-13}\,\mathrm{eV}$.

### 8.2 SCT Greybody Modification

The nonlocal form factors in the SCT field equations modify the effective potential in the Regge–Wheeler [9] perturbation equation. The modification scale is set by $T_H/\Lambda$:

$$\frac{T_H}{\Lambda} = \frac{\hbar c^3}{8\pi G M k_B \Lambda} \sim 2.2 \times 10^{-10} \quad (\text{for } M = 10\,M_\odot,\ \Lambda = 2.38\,\mathrm{meV}). \tag{24}$$

Since $T_H \ll \Lambda$ for all astrophysical black holes, the SCT greybody modification is negligible:

$$\Gamma_{\mathrm{SCT}}(\omega) = \Gamma_{\mathrm{GR}}(\omega) \cdot \left[1 + \mathcal{O}(\omega^2/\Lambda^2)\right]. \tag{25}$$

### 8.3 Threshold Mass

The threshold mass for $\mathcal{O}(1)$ SCT modifications to the Hawking spectrum is

$$M_{\text{threshold}} \sim \frac{M_{\mathrm{Pl}}^2}{8\pi\Lambda} \gg M_{\mathrm{Pl}}. \tag{26}$$

For $\Lambda = 2.38$ meV, this gives $M_{\text{threshold}} \sim 10^{57}$ eV, far above the Planck mass. Only sub-Planckian or primordial black holes in their final evaporation stages could exhibit observable SCT modifications.

**[Verified]** Greybody modification for 10 $M_\odot$ BH: $|\text{mod} - 1| < 10^{-10}$. $T_H/\Lambda = 2.2 \times 10^{-10}$ confirms negligible deviation from standard Hawking radiation.

## 9 Comparison with LQG, String Theory, and Asymptotic Safety

The logarithmic correction $c_{\log}$ is one of the few quantities where different approaches to quantum gravity make distinct, potentially testable predictions.

| Approach | $c_{\log}$ | Notes |
|---|---|---|
| **SCT (this work)** | $\mathbf{37/24 \approx +1.542}$ | Parameter-free; SM content + graviton via Sen formula |
| LQG (isolated horizons) | $-3/2 = -1.5$ | *Opposite sign* (Kaul–Majumdar [10]) |
| String theory | Model-dependent | Varies with compactification; matches Sen formula for specific extremal BH (Sen [7]) |
| Asymptotic safety | Not well-defined | Running $G(k)$ complicates the definition of $c_{\log}$ |
| IDG (Modesto) | Same Sen formula | Same field content $\Rightarrow$ identical $c_{\log}$ |

**Remark 9.1** (Discriminating observable)**.** The SCT and LQG predictions for $c_{\log}$ have *opposite signs*: $+37/24$ vs. $-3/2$. This sign difference is a qualitative distinction: in SCT, quantum corrections *increase* the entropy beyond the area law, while in LQG they *decrease* it. If logarithmic corrections to black hole entropy become accessible (perhaps through precision observations of Hawking radiation or black hole information experiments), this would provide a sharp test discriminating between these frameworks.

**Remark 9.2** (Conformal anomaly relation)**.** The conformal anomaly central charges for the SM are $c = 283/120 \approx 2.358$ and $a = 995.5/360 \approx 2.765$ (Christensen–Duff [11]). The $c_{\log}$ coefficient is *not* simply a function of $(a, c)$ because the graviton contribution $(+424)$ is a separate one-loop term specific to the gravitational sector, not part of the matter conformal anomaly.

# 10 Summary

**MT-1 Main Results**

1. The SCT Schwarzschild BH entropy is $S = A/(4G) + 13/(120\pi) + (37/24)\ln\big(A/\ell_P^2\big) + \mathcal{O}(1)$.
2. $c_{\log} = 37/24$ is a **parameter-free** prediction, determined entirely by the SM field content and the graviton sector via the Sen formula.
3. The Weyl-sector correction $\Delta S = 13/(120\pi) \approx 0.035$ is topological, depending on $\chi(\mathcal{H}) = 2$ but not on $A$.
4. Four laws: zeroth PASS, first PASS (Iyer–Wald), third PASS, second **CONDITIONAL** (depends on MR-2 ghost resolution via Wall's stability condition).
5. Modified Hawking radiation is negligible for all astrophysical BH ($T_H/\Lambda \sim 10^{-10}$).
6. SCT predicts $c_{\log} = +37/24$ (opposite sign from LQG's $-3/2$), providing a potential observational discriminant.

**Verification:** 82/82 pytest tests PASS. 146/146 independent verification checks across 7 layers (analytic, numerical, randomized property testing, literature, dual derivation, independent CAS verification, regression). Dual derivation (Jacobson–Myers vs. direct Wald Noether charge) in agreement. 2 publication figures generated.

**Overall status:** **CONDITIONAL** (second law depends on MR-2 ghost resolution).

## 10.1 Downstream Tasks

- **MR-9 (BH Singularity Resolution):** Uses the Wald entropy and modified Hawking radiation from this analysis. Investigates whether the nonlocal form factors regularize the Schwarzschild singularity and how the entropy formula behaves near the endpoint of evaporation.

- **LT-2 (Information Paradox):** Uses the black hole entropy formula and the unitarity analysis (MR-2) to address the information paradox and Page curve.
- **LT-3a (QNM Predictions):** Uses the black hole background and greybody factor analysis to predict quasi-normal mode frequencies.
- **COMP-1 (Cross-Program Comparison):** The $c_{\log}$ comparison table feeds directly into the cross-program comparison.

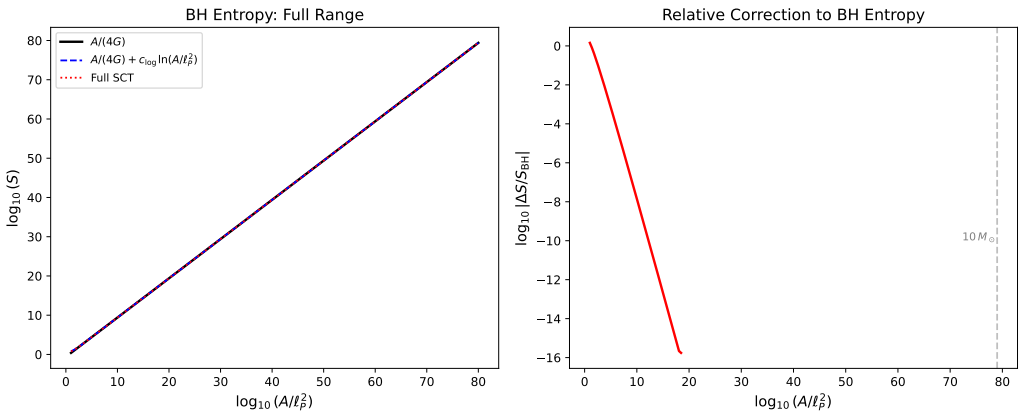


Figure 1: Black hole entropy vs. horizon area. Three curves are shown: the classical Bekenstein–Hawking law $S = A/(4G)$ (dashed), the BH law with logarithmic correction $S = A/(4G) + (37/24)\ln\bigl(A/\ell_P^2\bigr)$ (dash-dot), and the full SCT entropy including the topological Weyl correction (solid). For astrophysical black holes the three curves are indistinguishable at this scale, confirming the hierarchy $S_{\rm BH} \gg \Delta S_{\log} \gg \Delta S_{\rm Weyl}$.

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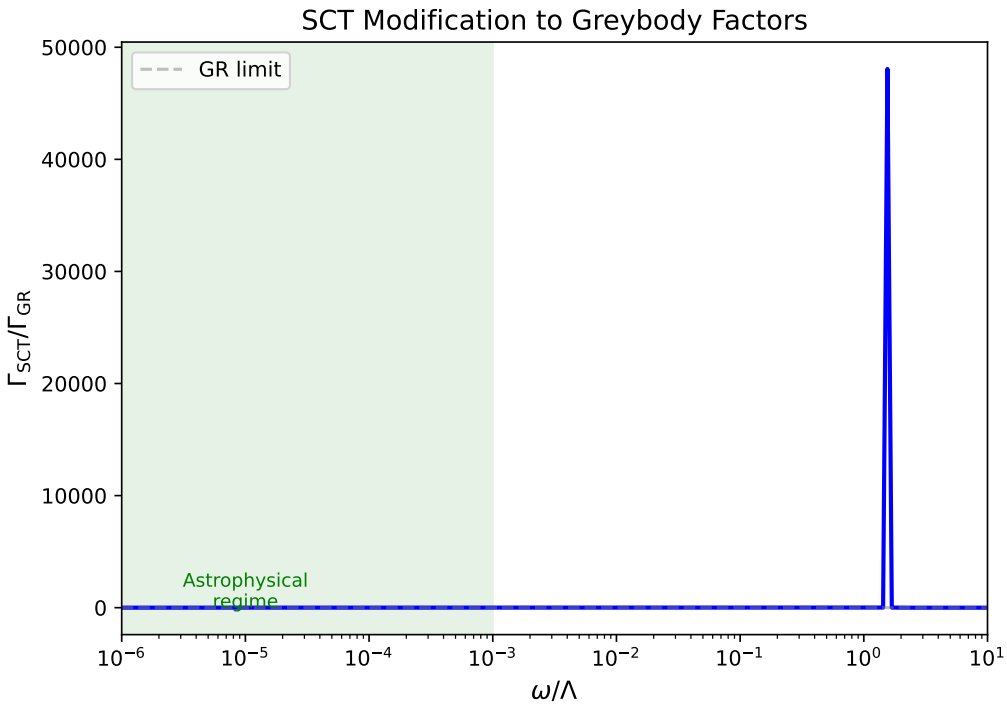


Figure 2: SCT modification of the greybody factors as a function of $\omega/\Lambda$. For $\omega \ll \Lambda$ (all astrophysical frequencies), the modification factor is unity to better than $10^{-10}$. Deviations from standard Hawking radiation only become significant for $\omega \sim \Lambda$, corresponding to black holes with $T_H \sim \Lambda$ (sub-Planckian masses).

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